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This Stereo Amplifier Is Simple To Make

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Here is a simple stereo amplifier you can make with a CD6283 audio amplifier and a few passive components. It provides an output of 4.6 watts per channel with a 12V power supply.

Circuit diagram of the stereo amplifier is shown in Fig. 1. CON1 is the connector for 6-12V input power supply, CON2 is for the input signal and CON3 provides the output to two speakers that may be connected to it.

Pins 5 and 7 of IC1 are connected to CON2 for audio inputs via capacitors C2 and C3, respectively. Pin 12 is connected to CON1 connector for power supply. Pin 2 of IC1 is connected to left (L) channel output via electrolytic capacitor C10 and pin 10 of IC1 is connected to right (R) channel output via electrolytic capacitor C9. A 4-ohm, 4-watt speaker each is connected at CON3 for left and right channels, respectively.

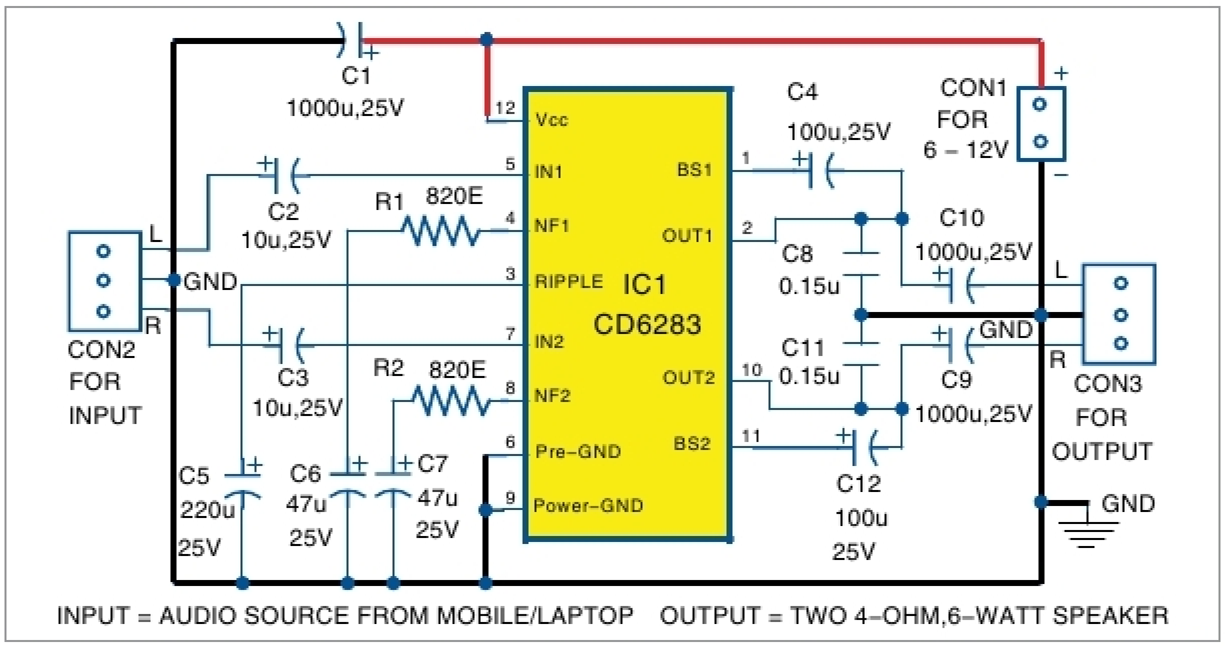

Fig. 1: Circuit diagram of the stereo amplifier

PARTS LIST	
Semiconductors:	
IC1	- CD6283 audio amplifier
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2	- 820-ohm
Capacitors:	
C1, C9, C10	- 1000µF, 25V electrolytic
C2, C3	- 10µF, 25V electrolytic
C4, C12	- 100µF, 25V electrolytic
C5	- 220µF, 25V electrolytic
C6, C7	- 47µF, 25V electrolytic
C8, C11	- 0.15µF ceramic disk
Miscellaneous:	
CON1	- 2-pin terminal connector
CON2, CON3	- 3-pin connector
	- 4-ohm, 4W speaker (2 Nos.)
	- Suitable heat-sink for IC CD6283
	- 6-12V DC regulated power supply

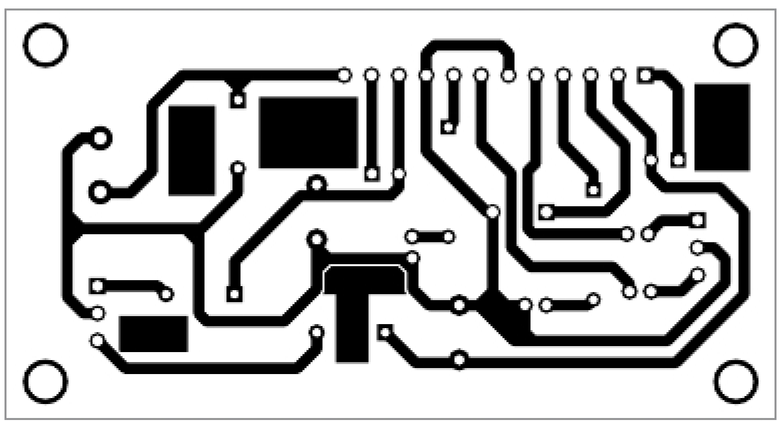

Fig. 2: Actual-size PCB layout for the stereo amplifier

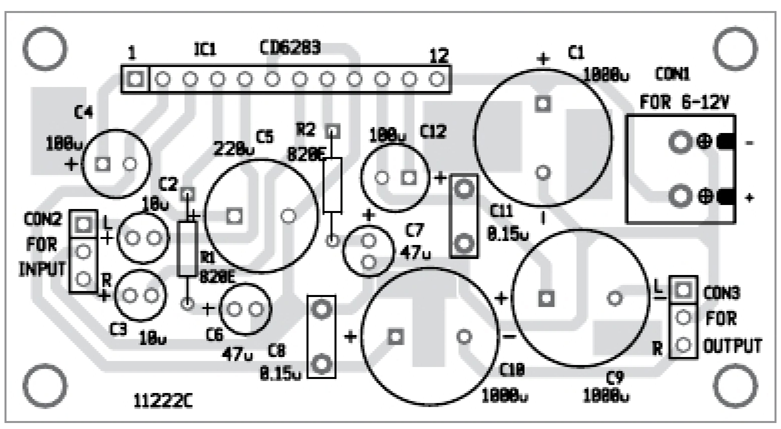

Fig. 3: Components layout for the PCB

Construction and testing

An actual-size PCB layout for the amplifier is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect a 6-12V power supply across CON1.

After completing the wiring, connect left (L) and right (R) stereo audio inputs from a mobile/laptop/desktop at CON2. Then connect the speakers at CON3. Take a 6V or 12V battery and connect it to the amplifier circuit.

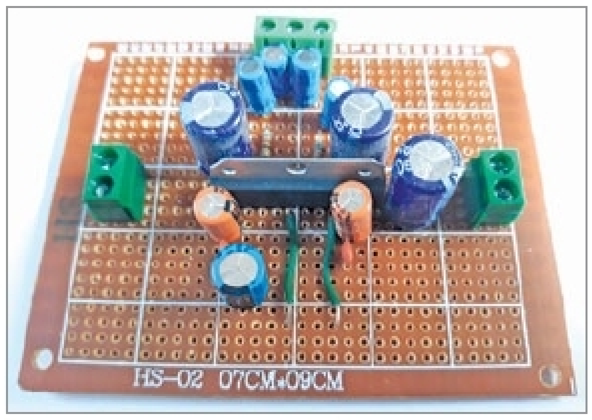

Fig. 4: Author's prototype

You can calibrate the circuit by connecting a signal generator (set at 1 kHz) to left and right inputs at CON2. You should hear a loud amplified tone from left and right speakers at the output. You can also use an MP3 player instead of signal generator to calibrate the circuit.

Optional 22-kilo-ohm potentiometers can be connected at CON2's left and right inputs for volume control. Your amplifier is ready to use! EFY

Raj K. Gorkhali is a hobbyist and a regular contributor to EFY